Conditional Probability and Bayes Rule

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

[Unit 3 · Probability, Random Variables & Distributions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

2 **[Foundational]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

A sample dataset has $Q_1 = 48$ and $Q_3 = 76$. Using the 1.5 IQR rule: (a) Determine the exact lower and upper outlier fences. (b) If the dataset includes values of 4 and 122, determine whether each is an outlier.

(a) Outlier Fences: Calculate $\text{IQR} = Q_3 - Q_1 = 76 - 48 = 28$. Lower fence: $Q_1 - 1.5(\text{IQR}) = 48 - 1.5(28) = 48 - 42 = \mathbf{6}$. Upper fence: $Q_3 + 1.5(\text{IQR}) = 76 + 1.5(28) = 76 + 42 = \mathbf{118}$. (b) Outlier Status: Since $4 < 6$, 4 is a low outlier. Since $122 > 118$, 122 is a high outlier.

3 **[Foundational]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x \to 0} \frac{\sqrt{1+\tan x} - \sqrt{1+\sin x}}{x^3}$.

Step 1 · Rationalize the Numerator: Multiply by $\sqrt{1+\tan x} + \sqrt{1+\sin x}$: $\lim_{x \to 0} \frac{(1+\tan x) - (1+\sin x)}{x^3(\sqrt{1+\tan x} + \sqrt{1+\sin x})} = \lim_{x \to 0} \frac{\tan x - \sin x}{x^3(\sqrt{1+\tan x} + \sqrt{1+\sin x})}$. Step 2 · Factor Trigonometric Terms: $\tan x - \sin x = \frac{\sin x}{\cos x} - \sin x = \sin x\left(\frac{1-\cos x}{\cos x}\right)$. Multiply numerator and denominator by $1 + \cos x$.

4 **[Foundational]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

A cereal manufacturer places a collectible card in each box: 20% of boxes have Card A, 30% have Card B, and 50% have Card C. Design a simulation using random digits to estimate the probability that a customer must buy at least 6 boxes to collect all three cards.

1. Assign Digits: Assign single digits proportional to probabilities: Digits 0–1 represent Card A (20%); digits 2–4 represent Card B (30%); digits 5–9 represent Card C (50%). 2. Describe One Trial: Read random digits one at a time. Record which card type is obtained on each digit. Continue reading digits until all three card types (A, B, and C) have appeared at least once. Count the total number of digits (boxes) required. 3. Record Statistic: Record whether the total number of boxes opened is ≥ 6 (Yes/No). 4..

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5 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Use Cramer's Rule to solve for x_2 in the linear system: $\{2x_1 + x_2 + x_3 = 4; x_1 - x_2 + 2x_3 = 1; 3x_1 + 2x_2 - x_3 = 7$

Step 1: Compute main determinant $\det(A)$: $\det(A) = |2, 1, 1; 1, -1, 2; 3, 2, -1| = 2((-1)(-1) - 2(2)) - 1(1(-1) - 2(3)) + 1(1(2) - (-1)(3))$ $= 2(1 - 4) - 1(-1 - 6) + 1(2 + 3) = 2(-3) - 1(-7) + 1(5) = -6 + 7 + 5 = 6.$ Step 2: Replace column 2 with $\mathbf{b} = (4, 1, 7)^T$: $\det(A_2(\mathbf{b})) = |2, 4, 1; 1, 1, 2; 3, 7, -1|.$ Expanding along row 1: $= 2(1(-1) - 2(7)) - 4(1(-1) - 2(3)) + 1(1(7) - 1(3)) = 2(-15) - 4(-7) + 1(4) = -30 + 28 + 4 = 2.$ Step 3: Evaluate x_2.

6 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A diagnostic medical screening test is 95% sensitive (positive if disease present) and 90% specific (negative if disease absent). In a population, 2% of people have the disease. Find the probability that a person who tests positive actually has the disease.

Let D = disease present, T^+ = positive test. $P(D) = 0.02, P(D^c) = 0.98.$ Sensitivity: $P(T^+ \mid D) = 0.95.$ Specificity: $P(T^- \mid D^c) = 0.90 \Rightarrow P(T^+ \mid D^c) = 0.10.$ By the Law of Total Probability: $P(T^+) = P(D)P(T^+ \mid D) + P(D^c)P(T^+ \mid D^c) = (0.02)(0.95) + (0.98)(0.10) = 0.019 + 0.098 = 0.117.$ By Bayes' Theorem: $P(D \mid T^+) = \frac{P(D \cap T^+)}{P(T^+)} = \frac{0.019}{0.117} \approx \mathbf{0.1624}$ (16.24%). Despite 95% sensitivity, only ~16.2% of positive testers.

7 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

Step 1 · Initial Verification: As $x \to 0$, numerator $0 - 0 = 0$ and denominator $0^3 = 0$. Form $\frac{0}{0}$. Differentiating numerator and denominator: $\lim_{x \to 0} \frac{1 - \frac{1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{\frac{(1+x^2)-1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{x^2}{3x^2(1+x^2)}.$ Step 2 · Algebraic Simplification: For $x \neq 0$, cancel the common factor x^2: $= \lim_{x \to 0} \frac{1}{3(1+x^2)} = \frac{1}{3(1+0)} = \mathbf{\frac{1}{3}}.$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 3 · AdvancedAn assembly line fills cereal boxes with a mean weight $\mu = 450$ grams and standard deviation $\sigma = 18$ grams. A random sample of $n = 36$ boxes is selected. What is the mean and standard error of the sample mean weight $\bar{x}$, and what is the probability that $\bar{x}$ exceeds 455 grams?

Sampling distribution parameters: $\mu_{\bar{x}} = \mu = \mathbf{450}$ g, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{18}{\sqrt{36}} = \frac{18}{6} = \mathbf{3}$ g. By the Central Limit Theorem ($n = 36 \geq 30$), $\bar{x}$ is approximately normally distributed: $z = \frac{455 - 450}{3} = \frac{5}{3} \approx 1.67.$ Evaluating tail probability: $P(\bar{x} > 455) = P(Z > 1.67) = 1 - 0.9525 = \mathbf{0.0475}$ (4.75%).

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9 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A discrete random variable X has probability distribution: $P(X=0)=0.1$, $P(X=1)=0.3$, $P(X=2)=0.4$, $P(X=3)=0.2$. Calculate: (a) $E(X)$, (b) $E(X^2)$, and (c) $\text{Var}(X)$ and σ_X.

(a) Expected Value: $E(X)=0(0.1)+1(0.3)+2(0.4)+3(0.2)=0+0.3+0.8+0.6=\mathbf{1.7}$. (b) Expected Value of X^2: $E(X^2)=0^2(0.1)+1^2(0.3)+2^2(0.4)+3^2(0.2)=0+0.3+1.6+1.8=\mathbf{3.7}$. (c) Variance & SD: $\text{Var}(X)=E(X^2)-[E(X)]^2=3.7-(1.7)^2=3.7-2.89=\mathbf{0.81}$. $\sigma_X=\sqrt{0.81}=\mathbf{0.9}$.

10 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A basketball player makes 75% of free throws. In a game, she shoots 12 free throws. Assuming attempts are independent: (a) Find the probability she makes exactly 9 free throws. (b) Find the mean and standard deviation of made free throws.

(a) Binomial PMF: $X\sim B(n=12,p=0.75)$, $k=9$: $P(X=9)=\binom{12}{9}(0.75)^9(0.25)^3=220\times0.075085\times0.015625\approx\mathbf{0.2581}$. (b) Mean & Standard Deviation: $\mu=np=12(0.75)=\mathbf{9.0}$ free throws.

$\sigma=\sqrt{np(1-p)}=\sqrt{12(0.75)(0.25)}=\sqrt{2.25}=\mathbf{1.5}$ free throws.

11 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 1 · FoundationsFind the derivative of $y=\sqrt{x^4+3x^2+7}$ using the Chain Rule, and compute $y'(1)$.

Express $y=(x^4+3x^2+7)^{1/2}$. By the Chain Rule: $y'=\frac{1}{2}(x^4+3x^2+7)^{-1/2}\cdot\frac{d}{dx}(x^4+3x^2+7)=\frac{4x^3+6x}{2\sqrt{x^4+3x^2+7}}=\frac{2x^3+3x}{\sqrt{x^4+3x^2+7}}$. Evaluate at $x=1$: $y'(1)=\frac{2(1)^3+3(1)}{\sqrt{1^4+3(1)^2+7}}=\frac{5}{\sqrt{11}}=\mathbf{\frac{5\sqrt{11}}{11}}$.

12 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 1 · FoundationsTwo independent events E and F satisfy $P(E)=0.35$ and $P(F)=0.60$. Find: (a) $P(E\cap F)$, (b) $P(E\cup F)$, and (c) $P(E\mid F)$.

(a) Intersection: Since E and F are independent: $P(E\cap F)=P(E)\cdot P(F)=0.35\times0.60=\mathbf{0.21}$. (b) Union: Using the General Addition Rule: $P(E\cup F)=P(E)+P(F)-P(E\cap F)=0.35+0.60-0.21=\mathbf{0.74}$. (c) Conditional: By definition of independence: $P(E\mid F)=P(E)=\mathbf{0.35}$.

13 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 2 · MediumLet X and Y be independent random variables with $\sigma_X=6$ and $\sigma_Y=8$. What is the standard deviation of the difference $D=X-Y$?

Variances ALWAYS add, even when subtracting random variables: $\text{Var}(X-Y)=\text{Var}(X)+\text{Var}(Y)=6^2+8^2=36+64=100$. The standard deviation is: $\sigma_{X-Y}=\sqrt{100}=\mathbf{10}$.

14 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

[Unit 2 · Collecting Data, Sampling & Experimental Design · Practice 4] If the governing relation for this system is $g(t)=6t+6$, evaluate $g(10)$ and find the inverse input t when $g(t)=72$.

Substitute $t=10$: $g(10)=6(10)+6=\mathbf{66}$. Setting $6t+6=72$ gives $6t=66\Rightarrow t=\mathbf{11}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

[Unit 4 · Statistical Inference: Proportions, Means & Slopes · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

16 **[Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS**

[Linear Transformations, Matrix Representations & Change of Basis · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 **[Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification**

[Orthogonality, Gram–Schmidt, QR Decomposition & Least Squares · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

18 **[Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

[Double Integrals, Fubini's Theorem, Polar Coordinates & The Change-of-Variables Jacobian · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 **[Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI**

[Green's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 **[Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

[Functions of Several Variables, Limits, Partial Derivatives & Differentiability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Consider the following diagram representing a probability space where events A and B are shown. If $P(A) = 0.6$ and $P(A \cap B) = 0.2$, and the area of the entire sample space is 1, what is the value of $P(B \setminus A)$?

a. 0.2

b. **0.4** ✓

c. 0.8

d. 0.6

The diagram shows overlapping events A and B within a sample space of area 1. We are given $P(A) = 0.6$ and $P(A \cap B) = 0.2$. We are asked to find $P(B \setminus A)$, which represents the probability of event B occurring but not event A. We know that $P(A) = P(A \cap B) + P(A \cap B^c)$. Since $P(A) = 0.6$ and $P(A \cap B) = 0.2$, we have $0.6 = 0.2 + P(A \cap B^c)$, which means $P(A \cap B^c) = 0.4$. Also, we know that the probability of event B can be expressed as $P(B) = P(A \cap B) + P(B \cap A^c)$. The nota.

22 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Consider a continuous random variable X with a uniform distribution over the interval $[2, 8]$. What is the probability that X is between 4 and 7, i.e., $P(4 < X < 7)$?

a. **0.50** ✓

b. 0.25

c. 0.75

d. 0.17

For a continuous uniform distribution over the interval $[a, b]$, the probability density function (PDF) is $f(x) = \frac{1}{b-a}$ for $a \le x \le b$, and 0 otherwise. In this case, $a = 2$ and $b = 8$, so the length of the interval is $b - a = 8 - 2 = 6$. The PDF is $f(x) = \frac{1}{6}$ for $2 \le x \le 8$. The probability $P(4 < X < 7)$ is the integral of the PDF from 4 to 7, or equivalently, the area under the PDF curve between 4 and 7. $P(4 < X < 7) = \int_4^7 \frac{1}{6} dx$ Since the PDF is constant, this is.

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23 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Consider a discrete random variable X that represents the number of heads obtained when a fair coin is tossed 3 times. The probability distribution of X is given by: | x | 0 | 1 | 2 | 3 | |---------|-------|-------|-------|-------| | $P(X = x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | What is the probability that X is less than 2, i.e., $P(X < 2)$?

a. $\frac{3}{8}$

b. $\frac{4}{8}$

c. $\frac{5}{8}$ ✓

d. $\frac{6}{8}$

The problem asks for the probability that the random variable X (number of heads in 3 coin tosses) is less than 2. This means we need to find $P(X < 2)$. The possible values of X that are less than 2 are $X = 0$ and $X = 1$. Therefore, $P(X < 2) = P(X = 0) + P(X = 1)$. From the given probability distribution: $P(X = 0) = \frac{1}{8}$ $P(X = 1) = \frac{3}{8}$ So, $P(X < 2) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8}$. Simplifying the fraction, $P(X < 2) = \frac{1}{2}$. Looking at the choices provided, $\frac{4}{8}$ is equival.

24 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A software company develops a new encryption algorithm. The algorithm's security depends on a sequence of random integers generated by a custom pseudo-random number generator (PRNG). The PRNG is designed such that the probability of generating an integer in a specific range $[a, b]$ is proportional to the length of the range, $b - a$, for $0 \le a \le b \le 1$. Let X be the first integer generated and Y be the second integer generated by this PRNG, independently. The problem is to find the probability that the sum of the first two generated integers, $S = X + Y$, is less than 1. Assume the PRNG generates numbers from a continuous uniform distribution on $[0, 1]$.

a. 1/4

b. **1/3** ✓

c. 1/2

d. 2/3

The problem states that the probability of generating an integer in a range $[a, b]$ is proportional to $b - a$ for $0 \le a \le b \le 1$. This describes a continuous uniform distribution on the interval $[0, 1]$. Therefore, the probability density function (PDF) for both X and Y is $f(x) = 1$ for $0 \le x \le 1$ and 0 otherwise. We are interested in the probability $P(X + Y < 1)$. Since X and Y are independent and both follow a $U(0, 1)$ distribution, their joint PDF is $f(x, y) = f(x)f(y) = 1$ for.

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25 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A quality control engineer is monitoring the number of defects produced by two different machines, Machine A and Machine B. Let X be the number of defects produced by Machine A in a given hour, and Y be the number of defects produced by Machine B in the same hour. The joint probability distribution of X and Y is given in the table below: | $P(X = x, Y = y)$ | $y = 0$ | $y = 1$ | $y = 2$ | | :----------: | :-------: | :-------: | :-------: | | $x = 0$ | 0.10 | 0.15 | 0.05 | | $x = 1$ | 0.20 | 0.30 | 0.10 | | $x = 2$ | 0.05 | 0.05 | 0.00 | What is the expected value of the total number of defects produced by both machines in a given hour, $E(X + Y)$?

a. 1.15

b. **1.30** ✓

c. 1.45

d. 1.60

To find the expected value of the total number of defects, $E(X + Y)$, we can use the property of linearity of expectation: $E(X + Y) = E(X) + E(Y)$. First, let's calculate the marginal probability distributions for X and Y. For X: $P(X = 0) = 0.10 + 0.15 + 0.05 = 0.30$ $P(X = 1) = 0.20 + 0.30 + 0.10 = 0.60$ $P(X = 2) = 0.05 + 0.05 + 0.00 = 0.10$ Now, calculate $E(X)$: $E(X) = (0 \times 0.30) + (1 \times 0.60) + (2 \times 0.10) = 0 + 0.60 + 0.20 = 0.80$ For Y: $P(Y = 0) = 0.10 + 0.20 + 0.05 = 0.35$.

26 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A discrete random variable, X, represents the number of successful trials in a sequence of independent Bernoulli trials. The probability of success on any given trial is $p = 0.6$. The expected value of X is $E[X] = 18$. We are interested in a new random variable, Y, defined as $Y = aX + b$, where a and b are constants. If the variance of Y is $Var(Y) = 7.2$, what is the value of a^2?

a. 0.16

b. **0.36** ✓

c. 1.00

d. 2.50

The random variable X follows a binomial distribution, as it represents the number of successes in a fixed number of independent Bernoulli trials. The expected value of a binomial random variable is given by $E[X] = np$, where n is the number of trials and p is the probability of success. We are given $E[X] = 18$ and $p = 0.6$. Therefore, $np = 18$ implies $n(0.6) = 18$. Solving for n, we get $n = \frac{18}{0.6} = 30$. So, there are 30 trials. The variance of a binomial random variable is given by.

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27 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A machine dispenses a liquid into containers. The amount of liquid dispensed, in milliliters, follows a continuous uniform distribution over the interval $[95, 105]$. A quality control inspector randomly selects 5 containers and measures the amount of liquid in each. Let X_i be the amount of liquid in the ith container, for $i = 1, 2, 3, 4, 5$. The company has implemented a new calibration process that, on average, increases the mean amount of liquid dispensed by δ milliliters, but also increases the variance by σ^2. The inspector is specifically looking for evidence that the calibration process has *decreased* the variability of the dispensed liquid while maintaining a satisfactory average fill. If the current calibration process yields a mean fill of $\mu_0 = 100$ ml and a standard deviation of $\sigma_0 = 2$ ml, and the new process is hypothesized to have a mean fill of $\mu_1 = 100 + \delta$ ml and a standard deviation of $\sigma_1 = 2 - \sigma_{dev}$ ml, where $\delta > 0$ and $\sigma_{dev} > 0$ are unknown positive constants, what is the minimum value of σ_{dev} such that the probability of falsely concluding that the new process has *increased* variability (i.e., Type I error) is at most 0.01, assuming the mean fill remains at 100 ml (i.e., $\delta = 0$)?

a. The probability cannot be determined without knowing the specific value of δ.

b. Approximately 1.01

c. **Approximately** 1.37 ✓

d. Approximately 2.00

Let X be the amount of liquid dispensed by the machine. X follows a continuous uniform distribution $U(95, 105)$. The mean is $\mu = \frac{95+105}{2} = 100$ ml. The variance is $\sigma^2 = \frac{(105-95)^2}{12} = \frac{100}{12} = \frac{25}{3}$. The standard deviation is $\sigma = \sqrt{\frac{25}{3}} = \frac{5}{\sqrt{3}} \approx 2.887$ ml. The problem statement implies the initial standard deviation is $\sigma_0 = 2$ ml, which contradicts the calculated standard deviation for a uniform distribution from 95 to 10.

28 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A bag contains marbles of different colors. The probability of drawing a red marble is $P(\text{Red}) = \frac{2}{5}$ and the probability of drawing a blue marble is $P(\text{Blue}) = \frac{1}{3}$. If the draws are independent, what is the probability of drawing a red marble followed by a blue marble?

a. $\frac{2}{15}$ ✓

b. $\frac{3}{8}$

c. $\frac{1}{2}$

d. $\frac{11}{15}$

Since the draws are independent, the probability of two independent events occurring in sequence is the product of their individual probabilities. Let R be the event of drawing a red marble and B be the event of drawing a blue marble. We are given: $P(R) = \frac{2}{5}$ $P(B) = \frac{1}{3}$ The probability of drawing a red marble followed by a blue marble is $P(R \text{ and } B) = P(R) \times P(B)$. $P(R \text{ and } B) = \frac{2}{5} \times \frac{1}{3} = \frac{2 \times 1}{5 \times 3} = \frac{2}{15}$. Therefo.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

A discrete random variable X can take on the values 1, 2, 3, and 4. The probability distribution for X is given by P(X=x) = kx, where k is a constant. What is the value of P(X=2)?

a. $\frac{1}{6}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{2}$

d. $\frac{5}{6}$

The random variable X can take on the values 1, 2, and 3. The probability distribution is given by $P(X = x) = kx$, where k is a constant. The sum of the probabilities for all possible values of X must equal 1. $P(X = 1) + P(X = 2) + P(X = 3) = 1$ Substituting the given formula for the probabilities: $k(1) + k(2) + k(3) = 1$ $k + 2k + 3k = 1$ $6k = 1$ Solving for k: $k = \frac{1}{6}$ Now we can find the probability $P(X = 2)$: $P(X = 2) = k(2)$ Substitute the value of k we found: $P(X = 2) = (\frac{1}{6})(2)$.

30 **[Intermediate]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

A discrete random variable X can take on the values $1, 2, 3,$ and 4. The probability mass function $P(X = x)$ is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. What is the value of k?

a. **1/10** ✓

b. 1/4

c. 1/6

d. 1/16

The sum of probabilities for all possible values of a discrete random variable must equal 1. Therefore, $\sum_{x=1}^{4} P(X = x) = 1$. Substituting the given probability mass function, we have: $P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 1$ $k(1) + k(2) + k(3) + k(4) = 1$ Factoring out k: $k(1 + 2 + 3 + 4) = 1$ $k(10) = 1$ Solving for k: $k = \frac{1}{10}$. Common algebraic traps include incorrectly summing the values of x or forgetting to set the sum of probabilities to 1. For example, setting $k = 1$ or $k = 1/4$ wo.

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31 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

a. 1/6

b. **1/3** ✓

c. 1/2

d. 2/3

The possible outcomes when rolling a fair six-sided die are {1, 2, 3, 4, 5, 6}. The total number of possible outcomes is 6. The outcomes that are greater than 4 are {5, 6}. The number of favorable outcomes is 2. The probability of an event is the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of rolling a number greater than 4 is $\frac{2}{6} = \frac{1}{3}$.

32 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Probability: $f(x, y) = 6xy$ on $0 \le x \le 1, 0 \le y \le \sqrt{1 - x^2}$ (quarter circle). $P(X + Y \le 1) =$

a. $\frac{3}{4}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$ ✓

d. $\frac{2}{3}$

Integrating over the triangle $x \ge 0, y \ge 0, x + y \le 1$ gives $\int_0^1 \int_0^{1-x} 6xy \, dy \, dx = 3\int_0^1 x(1-x)^2 \, dx = 3\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) = \frac{1}{4}$.

33 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Expected value E(X) for discrete X:

a. The most frequent value

b. $\sum x_i P(x_i)$ ✓

c. The median

d. $n \cdot p$ always

Applying Unit 3 · Probability, Random Variables & Distributions principles, Weighted average of outcomes. Therefore, the verified correct option is $\sum x_i P(x_i)$.

34 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

If P(A)=0.4 and P(B|A)=0.5, then P(A∩B)=

a. **0.2** ✓

b. 0.5

c. 0.4

d. 0.9

Applying Unit 3 · Probability, Random Variables & Distributions principles, P(A∩B)=P(A)P(B|A)=0.4×0.5=0.2. Therefore, the verified correct option is 0.2.

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35 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A binomial random variable has $n = 25$ and $p = 0.64$. Find its mean $\mu = np$.

a. 13

b. 14

c. **16** ✓

d. 18

Applying Unit 3 · Probability, Random Variables & Distributions principles, $\mu = np = 25 \times 0.64 = 16$. Therefore, the verified correct option is 16.

36 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

If $P(A) = 0.56$, find $P(A^c)$ (the complement of A).

a. **0.44** ✓

b. 0.54

c. 0.64

d. 0.24

Applying Unit 3 · Probability, Random Variables & Distributions principles, $P(A^c) = 1 - P(A) = 1 - 0.56 = 0.44$. Therefore, the verified correct option is 0.44.

37 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A geometric random variable has success probability $p = 0.48$. Find the mean (expected number of trials until first success), $\mu = 1/p$.

a. **2.08** ✓

b. 8.08

c. 4.08

d. 5.08

Applying Unit 3 · Probability, Random Variables & Distributions principles, $\mu = 1/p = 1/0.48 = 2.08$. Therefore, the verified correct option is 2.08.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A data set has $Q_1 = 17$ and $Q_3 = 35$. Using the 1.5×IQR rule, which of the following values would be classified as an outlier?

a. 35

b. **-14** ✓

c. 17

d. -9

IQR $= 35 - 17 = 18$. Lower fence $= 17 - 1.5(18) = -10$. Upper fence $= 35 + 1.5(18) = 62$. The value -14 is below the lower fence, so it is an outlier.

39 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Standard deviation of Binomial(n,p) is:

a. np

b. $np(1-p)$

c. $\sqrt{np(1-p)}$ ✓

d. $\sqrt{np}$

Applying Unit 3 · Probability, Random Variables & Distributions principles, σ=√(np(1-p)). Therefore, the verified correct option is $\sqrt{np(1-p)}$.

40 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Probability of event $a \le X \le b$ for PDF f:

a. $\int_a^b f(x)\,dx$ ✓

b. $f(b) - f(a)$

c. $f(a) \cdot (b - a)$

d. $F(b) \cdot F(a)$

Applying Applications of Integration principles, Probability = integral of PDF. Therefore, the verified correct option is $\int_a^b f(x)\,dx$.

41 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Error bound of Simpson's rule on $[a, b]$ with n intervals:

a. $O(h^2)$

b. $O(h^5)$

c. $O(h^3)$

d. $O(h^4)$ ✓

Applying Applications of Integration principles, Simpson: error $= O(h^4)$. Therefore, the verified correct option is $O(h^4)$.

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42 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Using L'Hôpital's rule: $\lim_{x \to 0} \frac{\sin 3x}{5x} =$

a. 3/5 ✓

b. 5/3

c. 0

d. 1

Applying Limits and Continuity principles, $\cos 3x \cdot 3/(5) \to 3/5$. Therefore, the verified correct option is 3/5.

43 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Probability density: $\int_{-\infty}^{\infty} f(x)\,dx =$

a. 0

b. ∞

c. 1 ✓

d. 1/2

Applying Applications of Integration principles, Total probability = 1. Therefore, the verified correct option is 1.

44 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

For $f(x, y) = 8x - 15y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}} f$ at any point, over all unit vectors $\mathbf{u}$.

a. 23

b. 16

c. 18

d. **17** ✓

$ablaf = (8, -15)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $ablaf = (8)^2 + (-15)^2 = 17$.

45 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

For $z = -3xy$ with $x = t^2$, $y = t$, find dz/dt at $t = 3$ using the chain rule.

a. -27

b. 81

c. **-81** ✓

d. -78

$z_x = -3y$, $z_y = -3x$, $x'(t) = 2t$, $y'(t) = 1$. $dz/dt = z_x x' + z_y y' = -9t^2$, so at $t = 3$: $dz/dt = -81$.

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